

Evaluating Privacy Loss in Kernel-Level Anti-Cheat Systems

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Abstract

This paper examines the ethical tradeoff between user privacy and competitive integrity in kernel-level anti-cheat systems used in modern online games. As cheating methods have become increasingly sophisticated, some anti-cheat providers have adopted Ring 0 access to detect low-level manipulation, direct memory access attacks, and hardware spoofing. However, this deeper system access creates privacy, transparency, and device-control concerns for users. This paper proposes a privacy loss scoring framework based on five categories: privilege level, runtime behavior, data collection risk, transparency, and device control. The framework is applied to four major anti-cheat systems: Riot Vanguard, BattlEye, Easy Anti-Cheat, and Activision Ricochet. The analysis finds that Riot Vanguard presents the highest privacy loss due primarily to its boot-time kernel driver and strong device-control model, while Easy Anti-Cheat presents the lowest overall privacy loss because of its more standardized and legally structured approach. The paper argues that kernel-level anti-cheat systems represent a difficult ethical balance between protecting fair competition and preserving user autonomy, transparency, and privacy.

1 Introduction

The data science project I have chosen examines kernel-level anti-cheat systems in modern video games. Today, the video game industry generates billions of dollars in revenue and attracts massive audiences across streaming platforms and social media. As the industry's cultural and economic influence has grown, so have the incentives for malicious actors to cheat, exploit competitive systems, and gain attention, status, or financial benefit.

Over time, cheating methods have become increasingly advanced. Some modern cheats use techniques such as direct memory access, allowing them to interact with hardware in ways that can bypass traditional anti-cheat systems operating at the user level. In response, some anti-cheat providers have moved deeper into the operating system, using Ring 0, or kernel-level, access to detect and prevent these threats. This leads to the central ethical issue of my project: the tradeoff between user privacy and competitive integrity. Kernel-level anti-cheat may help protect fair play, but it also introduces new risks because of the depth of access these systems have to a user's device. To evaluate this tradeoff, I propose a privacy loss score: a combined quantitative and qualitative measure of the privacy risk introduced by four major anti-cheat systems.

2 Background

Early anti-cheat systems were often designed to detect suspicious behavior from the same general level of access as ordinary software. Maario et al. (2021) discuss the risks associated with kernel-level anti-cheat systems in online gaming. Greidanus (2017) examines client-side anti-cheat systems from a privacy and data protection perspective. These user-level systems could scan for known cheat programs, detect unusual game files, monitor certain processes, or identify abnormal player behavior. However, as anti-cheat systems

improved, cheat developers also adapted. More advanced cheats began using techniques that made them harder to detect, including hiding processes, manipulating memory, and spoofing input. This created a so called arms race between cheat developers and anti-cheat providers.

Kernel-level anti-cheat emerged as a response to this arms race. The kernel is the core part of the operating system, responsible for managing hardware, memory, processes, and communication between software and the machine itself. Software that runs at the kernel level, often described as Ring 0 access, has far deeper privileges than normal applications running in user mode. From a security perspective, this deeper access can help anti-cheat systems detect hidden drivers, suspicious memory manipulation, unauthorized hardware interactions, and other methods that may be invisible to ordinary software. In other words, kernel-level access gives anti-cheat systems a better chance of seeing the same low-level activity that sophisticated cheats are trying to exploit.

3 Methodology

This project uses five categories to calculate the privacy loss score: privilege level, runtime behavior, data collection risk, transparency, and device control. These categories were chosen because they capture a wide range of behaviors that affect user privacy, trust, and system security. Each category examines a different part of the relationship between the anti-cheat provider and the user. Together they create a more complete picture of how much privacy risk each anti-cheat system introduces. Dornier and Klausner (2024) argue that some kernel-level anti-cheat systems share characteristics with rootkits. Privilege level matters because it determines how deeply the software can access a user's personal computer. A program that runs at the user level has limited access compared to a program that operates at the kernel level. Kernel-level access gives anti cheat systems more power to detect advanced cheating methods, but it also increases the potential consequences if the system is misused, poorly secured, or compromised. Runtime behavior is important because when the anti-cheat runs affects both user trust and privacy exposure. Some systems only run while the game is open, while others may start when the computer boots. The longer an anti-cheat system is active, the greater the concern about what activity it can observe, what data it may collect, and what security risks it may introduce.

Data collection focuses on what information is actually gathered, stored, transmitted, or analyzed. An anti-cheat system that collects limited data only when suspicious behavior is detected creates a different level of risk than one that continuously monitors system activity or sends detailed information to company servers. Transparency is important because users are being asked to trust software that may operate at a very deep level of their device. Clear communication from developers can help users understand what the anti-cheat does, when it runs, what data it collects, and why certain design choices were made. Without transparency, users may

Table 1: Privacy Loss Scoring Categories

Category	Score Range	Description
Privilege Level	0–20	Measures how deeply the anti-cheat system operates within the user’s device.
Runtime Behavior	0–20	Measures when the anti-cheat system runs, such as only during gameplay or at system startup.
Data Collection Risk	0–20	Measures the amount, sensitivity, transmission, and retention of user data.
Transparency	0–20	Measures how clearly the provider explains system behavior, data use, and enforcement decisions.
Device Control	0–20	Measures the severity of hardware bans, device fingerprinting, and repeat-offender controls.

feel that they are giving up privacy without fully understanding the risks or benefits.

Device control focuses on how the system responds to cheating or suspicious behavior. This includes bans, hardware identifiers, account restrictions, and attempts to prevent repeat offenders from returning. These controls can be effective for protecting competitive integrity, but they can also raise ethical concerns if users are punished unfairly, if appeals are unclear, or if device-level restrictions affect people beyond the original offender.

4 Four Major Anti Cheat Systems

The four major anti-cheat systems examined in this project are Riot Vanguard, BattlEye, Easy Anti-Cheat, and Activision Ricochet. These systems were chosen because they represent some of the most visible and widely discussed approaches to modern anti cheat enforcement. Riot Vanguard is used by Riot Games to protect competitive titles such as Valorant and League of Legends. Activision’s Ricochet Anti-Cheat is primarily associated with the Call of Duty franchise. BattlEye and Easy Anti-Cheat differ slightly because they are licensed to many different game developers and appear across a wide range of titles. Together, these four systems provide a useful comparison because they show how different companies approach the same core problem: protecting competitive integrity while managing the privacy and security concerns created by increasingly powerful anti-cheat software.

5 Privacy Loss Categories

5.1 Privilege

All four operate at Ring 0, so each system received the maximum score of 20 in this category.

5.2 Runtime

Riot Vanguard’s kernel driver starts when the system boots, while the client starts when the game launches. The other three systems only start their kernel drivers when the game is launched.

5.3 Data Collection Risk

Riot Vanguard raises a significant data collection concern because its kernel driver runs in the background after system startup. Riot has stated that the kernel driver does not have network connectivity.

Instead, it functions as a passive watchdog that communicates with the game client when the game starts to determine whether the system environment can be trusted.

Easy Anti Cheat is only active when the game is running. However EAC has telemetry data streaming to the backend for real time analysis. If nothing malicious is found the data is discarded within five days. When there is suspicious activity found that session’s data is flagged and stored for manual analysis. If the suspicious activity is determined to be a false alarm, the data is discarded within one to three months. If the data is a known cheat pattern the evidence is stored indefinitely.

BattlEye operates under a data minimization policy. For normal gameplay the system scans locally in real time and if nothing suspicious is found they do not store or transmit data to their servers. If suspicious behavior is found they store evidence for as long as they provide anti cheat services for that specific game.

Activision Ricochet is a multilayered system. When booting up the game there is an initial check with the TPM and secure boot state and is passed to their remote servers to attest that the PC is safe. During active gameplay, data is continuously sent to Ricochet servers for analysis. The system relies heavily on server-side machine learning to detect unnatural gameplay patterns. If the data is clean, then they discard it quickly. Suspicious gameplay usually takes around 3 – 14 days to review, so that data is retained at least that long. There is no public information about how long they keep the data.

5.4 Transparency

Riot’s engineering team communicates extensively about how and why it uses this type of anti-cheat system. They publish extensive, highly technical engineering blogs detailing the architecture of the systems. They published articles about why Ring 0 access was necessary to combat modern Direct Memory Access hardware cheats. One could define this approach as justification through education.

Activision’s Ricochet team frequently publishes anti-cheat progress reports that announce large scale ban numbers. They do not release the algorithms that are used in the process of banning these players. This creates a “black box” problem common in modern machine learning systems, making it difficult to explain which weighted feature or combination of inputs crossed the threshold into a ban.

Easy Anti Cheat is licensed out to hundreds of different game studios, so their transparency is rigid, standardized and entirely focused on legal compliance. Its policies provide clear information about data retention and GDPR compliance. However, they offer very little technical transparency to the end user. EAC represents transparency as a legal checklist rather than a moral obligation. They tell you exactly what data categories they collect to satisfy privacy law.

BattlEye appears to be the most opaque of the four systems based on the amount of public-facing technical information available. It is managed by a much smaller, highly secretive team compared to the larger corporate actors such as Epic Games and Riot Games. They rarely publish technical blogs, do not explain methodologies, and operate almost exclusively behind the scenes. Their philosophy appears to be that giving users too much information creates a security risk. If a player is banned, the message may be limited to a vague error code or a brief explanation, which can make the appeals process feel unclear.

5.5 Device Control

All four systems use hardware ID bans for repeat or severe offenders. A hardware ID is a fingerprint generated by combining unique identifiers from a computer's components. Because the anti-cheats run at kernel level they can query the hardware directly for these numbers.

Riot Vanguard appears to use one of the strongest device-control models among the four systems examined. They have shifted towards a hardware rooted trust. On Windows 11, Vanguard requires Trusted Platform Module 2.0, also known as TPM 2.0, and Secure Boot to be enabled. This allows Vanguard to tie a ban not just to a disk serial number, but to the unique cryptographic keys embedded in the security chip on your motherboard. Traditional hardware IDs can sometimes be masked by software spoofer. By contrast, bypassing a TPM-level ban may require physically replacing the motherboard or CPU. Ricochet goes beyond just banning the PC, it focuses on Input Device Control. In Call of Duty games, one major issue involves hardware spoofing devices, such as Cronus or XIM, that can give controller users mouse-and-keyboard-like precision or scripted inputs. They detect these problems with peripheral mapping called mousetrap. They try to fingerprint the behavior of the hardware, not just ban the hardware.

Easy Anti Cheat and BattlEye use the more traditional method of fingerprinting the hardware unique identifiers. Because these anti-cheat systems are licensed to many different game developers, their device-control policies are more game-specific.

6 Limitations

This analysis relies on publicly available documentation, privacy policies, developer blog posts, and secondary technical analysis. Because anti-cheat systems are intentionally opaque, some implementation details may be incomplete or unavailable. The privacy loss score should therefore be interpreted as a comparative ethical framework rather than a definitive technical audit. Additionally, the framework emphasizes privacy risk and does not directly measure anti-cheat effectiveness, false positive rates, or actual security outcomes.

7 Conclusion

To conclude, Riot Vanguard received the highest privacy loss score in this analysis. The main factors that separated Vanguard from the other systems were its runtime behavior and its lower transparency score. Unlike the other three systems, Vanguard's kernel driver starts when the computer boots, which increases the amount of time the anti cheat is active on the user's device. This creates a greater privacy concern even if the system is designed for cheat detection rather than general surveillance. The lowest overall privacy loss score was assigned to Easy Anti-Cheat. Its role as a large corporate provider that licenses anti-cheat software to many different developers appears to encourage a more standardized and risk-minimizing approach. This was reflected in its consistent score of 10 across the four categories other than privilege level.

However, all four systems face the difficult task of detecting advanced cheats in competitive gaming environments. Cheat developers have strong financial incentives to keep adapting, which forces anti-cheat providers to constantly respond with new detection methods. For that reason, the ongoing adversarial cycle between cheat developers and anti-cheat systems is unlikely to end soon. This project attempted to quantify the privacy risks involved and show how each system balances competitive integrity against user privacy.

Table 2: Privacy Loss Scores by Anti-Cheat System

System	Priv.	Run.	Data	Trans.	Device	Total
Riot Vanguard	20	20	10	5	15	70
BattlEye	20	10	5	15	15	65
Easy Anti-Cheat	20	10	10	10	10	60
Ricochet	20	10	15	10	10	65